Harvest Preservation Field Manual

12 methods to preserve your harvest from season to season

$16

Open-pollinated · Community-grown · Food sovereignty for everyone

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Seedwarden | $16

A commercial food system preserves food with infrastructure you can't replicate at home: continuous refrigeration chains, industrial retorts, modified-atmosphere packaging, nitrogen flushing, chemical dipping. You don't need any of that.

What you need is physics. Salt draws out water. Acid stops microbial growth. Heat destroys pathogens. Cold slows everything down. Dehydration removes the water microorganisms need to survive. Sugar binds available water. Alcohol poisons cellular machinery. These are the mechanisms that kept every human community fed through the winter long before there were freezers or USDA guidelines.

This guide covers twelve preservation methods. Each one is something an individual can do at home, with equipment that has a one-time cost and skills that, once learned, you keep. Some of them are nearly free (root cellaring, salt brining). Some require modest investment (a dehydrator, a pressure canner). All of them reduce your dependence on a food system that assumes you'll buy new food every week.

The goal is to grow in spring and summer, preserve through fall, and eat through winter — the way every agricultural community operated for ten thousand years before centralized food processing made that knowledge feel unnecessary.

How to Use This Guide

Each method section covers what it works for, what it doesn't, the complete equipment you need, step-by-step process with specific temperatures and ratios, critical safety information, how long your preserved food will last, and how to recognize spoilage.

Start with your harvest, not the method. The crop-to-method matrix at the back of this guide tells you which methods work for each crop. Match your harvest to your available equipment and choose from there.

On safety: Preservation methods that involve heat, anaerobic environments, or long-term storage have real failure modes. Botulism from improperly canned food is rare but serious. Every safety warning in this guide is based on tested science, not excessive caution. Read them.

What's covered here vs. the companion guides:

- Fermented & Preserved Harvest Handbook (Seedwarden): goes deeper on small-batch lacto-fermentation for apartment gardeners — techniques, troubleshooting, specific recipes. If fermentation is your primary method, that guide is worth reading alongside this one.
- This guide covers all twelve methods at full harvest scale, including pressure canning, root cellaring, smoking, and salt curing, which aren't covered in the companion handbook.

Important Notice

Food preservation has real failure modes, and some of them — particularly botulism from improperly canned food — can be fatal without any visible warning. pH testing is mandatory for water bath canning; if you do not test, you are guessing with your health. Follow tested recipes exactly as written, especially processing times, acid ratios, and pressure levels. Do not substitute ingredients, skip steps, or improvise until you fully understand the science involved. When in doubt about any preserved food, throw it out. No jar of food is worth a trip to the emergency room.

Part One: Equipment

Before you start preserving, know what you have and what you need. Equipment is organized here by entry level — you don't need to buy everything at once.

Level 1: What You Can Start With Today

These methods require only basic kitchen equipment and minimal expense. If you have a kitchen, you can start here.

For root cellaring and cold storage:

- Cool basement corner, garage, or unheated room with exterior wall
- Cardboard boxes or plastic bins with ventilation holes
- Dry sand, clean sawdust, or shredded leaves
- Thermometer and humidity gauge (hygrometer) — these matter; guessing doesn't work

For lacto-fermentation:

- Wide-mouth quart and half-gallon Mason jars
- Kitchen scale (mandatory — salt ratios must be accurate)
- Non-iodized pickling salt or kosher salt
- Small zip-lock bags filled with brine (as weights to keep vegetables submerged)

For basic freezing:

- An upright or chest freezer maintained at 0°F or below
- Heavy-duty zip-lock freezer bags (gallon and quart)
- Permanent marker and masking tape for labeling
- Large pot for blanching (8+ quart)
- Colander and large mixing bowl
- Digital instant-read thermometer

Level 2: Canning, Pickling, and Dehydrating

One-time investments that open up shelf-stable preservation.

- Boiling water canner — large enameled or stainless pot, 16+ quart, with jar rack. This handles all high-acid water bath canning.
- Mason-style canning jars — Ball, Kerr, Bernardin, or Golden Harvest only. Do not use commercial pasta sauce jars, mayo jars, or pickle jars. Buy a mixed case of half-pint (8 oz), pint (16 oz), and quart (32 oz) sizes. Wide-mouth for large pieces; regular mouth for liquids, jams, and purees.
- Two-piece lids — flat sealing discs (new every time; do not reuse) and metal bands (reusable if undamaged and rust-free).
- Jar lifter — required; hot jars cannot be moved safely by hand.
- Canning funnel — reduces mess and contamination significantly.
- Bubble remover / headspace tool — thin rigid spatula; a standard butter knife works.
- Candy / jelly thermometer — for jam, jelly, and low-temperature pasteurization.
- Food dehydrator — 4–9 tray, thermostat range 85–160°F, built-in fan. Horizontal airflow is preferable to vertical (reduces flavor transfer between trays). Nesco, Excalibur, and Cosori are reliable mid-range options.
- Bottled lemon juice — use bottled, not fresh; consistent acid level is required for tomato acidification. Keep several bottles on hand at harvest time.
- Citric acid powder — a more concentrated alternative to lemon juice for acidification; also useful in anti-browning dips.
- Powdered pectin — Sure-Jell or Ball brand, for jams and jellies.
- Ascorbic acid (Fruit-Fresh or pure vitamin C powder) — anti-browning treatment for cut fruit before drying or canning.

Level 3: Pressure Canning and Curing

For preserving low-acid vegetables, beans, meat, and cured products.

- Pressure canner — not a pressure cooker. Minimum 16-quart capacity. Two reliable types:

- Presto 23-quart (dial gauge) — requires annual calibration for gauge accuracy; bring to your county extension office each spring.

- All American 21.5-quart or 25-quart (weighted gauge) — self-regulating jigger weight; no calibration needed; more expensive but maintenance-free.

- Vacuum sealer with bags — extends dehydrated and frozen food quality significantly. Not required, but a meaningful upgrade.
- pH meter or pH strips — for verifying acid level when adapting tomato or salsa recipes. Required if you're working with unusual or low-acid varieties.
- Curing salts (Prague Powder #1 / Cure #1) — for curing pork and other meats. Contains 6.25% sodium nitrite. Required for any meat cured for long-term storage or smoked at low temperatures (prevents botulism in anaerobic, protein-rich environments).

- Smoker — electric, pellet, or offset barrel smoker. Used in combination with salt curing for long-term smoked preservation.
- Probe thermometer — leave-in type for monitoring internal temperature during smoking.

Part Two: Methods

Method 1: Root Cellaring and Cold Storage

Root cellaring is the simplest form of long-term preservation and requires no processing at all. The goal is to create conditions that slow metabolic activity and microbial growth without stopping them entirely. Cold + humidity control + good air circulation is the whole system.

What it works for:
Potatoes, sweet potatoes, winter squash, onions, garlic, carrots, beets, turnips, parsnips, rutabaga, celeriac, kohlrabi, cabbage, apples, pears.

What it does not work for:
Tomatoes, cucumbers, summer squash, peppers, most leafy greens, soft fruits, anything that can't tolerate cold.

The Four Storage Environments

Different crops need different conditions. Getting this wrong is one of the most common root cellaring mistakes.

Environment	Temperature	Humidity	Crops
Cold and moist	32–40°F	90–95% RH	Carrots, beets, turnips, parsnips, celer
Cold and dry	32–40°F	60–70% RH	Onions, shallots (after full curing)
Cool and moist	50–60°F	85–90% RH	Potatoes, cabbage
Warm and dry	50–60°F	60–70% RH	Winter squash, pumpkins, sweet potatoes,

A thermometer and humidity gauge (hygrometer) are not optional. Guessing will get you rotting vegetables.

Crop-Specific Instructions

Potatoes: Cure first at 50–60°F for 10–14 days to harden skins and heal cuts made during harvest. After curing, move to 38–40°F at 85–90% humidity. Store in complete darkness — light causes greening and solanine accumulation (toxic in large quantities). Remove all green-tinged potatoes. Do not store near apples (ethylene gas from apples causes sprouting). Stores **4–6 months**.

Winter squash and pumpkins: Cure at 80–85°F for 10–14 days before storage. Do not cure acorn squash (it overcures and develops off-flavors). After curing, store at 50–55°F at 60–70% humidity. Butternut keeps up to 6 months; acorn and delicata typically 2–3 months. Handle gently — bruises rot quickly.

Sweet potatoes: Cure at 80–85°F for 7–10 days. After curing, store at 55–60°F. Do not let them drop below 55°F — chilling injury causes hard cores and off-flavors. In Houston and other warm-climate areas, a climate-controlled interior room at 55–60°F is the correct storage location. Stores **4–6 months** after curing.

Onions: Spread in single layers in a warm (80–90°F), dry, well-ventilated space for 2–4 weeks until skins are completely papery and necks are fully dry. Any onion with a green neck or soft spot goes in the kitchen first, not storage. After curing, store at 32–40°F, 60–70% humidity, in mesh bags or open crates. Long-keeping storage varieties (Copra, Yellow Sweet Spanish) last **6–12 months**. Short-day sweet onions (Vidalia) don't store well — use within 1–2 months.

Garlic: Cure the same way as onions — warm, dry, ventilated for 2–4 weeks until wrappers are papery. Store at 32–40°F (hardneck types) or at cool room temperature (55–65°F works for softneck types). Good hardneck garlic stores **6–8 months**; softneck stores up to 12.

Carrots: Remove tops (leave ½ inch stub — cutting flush encourages root-end rot). Do not wash before storage. Pack in boxes with damp sand, clean sawdust, or shredded leaves. Store at 32–40°F, 90–95% humidity. Alternatively, mulch heavily in the ground and leave until needed — carrots sweeten after frost. Stores **4–6 months**.

Cabbage: Leave outer wrapper leaves on — they protect the interior head. Store at 32–40°F, 90–95% humidity. The smell will permeate anything nearby. Stores **3–4 months** if heads were cut when fully mature and immediately before first hard freeze.

Apples: Store at 32–35°F, 90–95% humidity. Keep away from all vegetables — apples produce ethylene gas that causes premature ripening and sprouting. Check weekly and remove any apple showing soft spots immediately. Different varieties store differently: Winesap, Northern Spy, and Granny Smith keep 3–5 months; Fuji 3–5 months; Red Delicious 3–5 months. Early summer varieties (Lodi, Yellow Transparent) don't store — process or eat immediately.

Signs of Spoilage

- Soft spots, shriveling, or sunken areas — remove affected items immediately
- Any fuzzy mold growth — gray, white, or black
- Rotting or fermented odor
- For potatoes: green patches (peel deeply; discard if extensive), hollow center, slimy surface
- For onions and garlic: sprouts emerging (still edible if caught early, but use soon), slimy neck, black powder mold (Aspergillus niger — discard)

Method 2: Freezing

Freezing is the most accessible and lowest-risk preservation method for most home growers. It doesn't require specialized knowledge, the equipment is already in most homes, and virtually everything can be frozen. The tradeoffs: electricity dependency and the fact that frozen food quality degrades over time even at safe temperatures.

What it works for:
Virtually all vegetables (with blanching), most fruits, meats, poultry, fish, cooked beans, pesto, tomato sauce, corn, peas, berries.

What it does not work for:
Whole eggs in shell, lettuce, cucumbers, radishes and other raw crisp vegetables (become waterlogged), cream sauces (separate), fried foods (lose crispness), fresh basil (turns black — infuse in oil first), potatoes frozen raw (become grainy and mealy — must blanch or cook first).

Blanching: Why It Matters

Blanching is briefly boiling or steaming vegetables before freezing. It deactivates the enzymes that continue breaking down cell structure, color, and flavor even at freezer temperatures. Skipping blanching doesn't make frozen vegetables unsafe — it makes them taste stale within 1–3 months rather than remaining good for 8–12 months.

The only vegetables that don't need blanching before freezing: peppers (freeze raw), onions (freeze raw for cooking use), fresh herbs (freeze raw or in oil), corn on the cob (blanch before cutting).

Blanching process:

1. Bring a large pot of water to a full rolling boil — use at least 1 gallon of water per pound of vegetable
2. Add vegetables; begin timing when water returns to a full boil (not when you drop the vegetables in)
3. After the specified time, immediately transfer to a large bowl of ice water for the same duration
4. Drain thoroughly and pat dry before packing — excess water creates ice crystals that degrade texture

Blanching times:

Vegetable	Water Blanch Time	Notes
Asparagus (thin)	2 min	Thick stalks: 4 min
Green/snap beans	3 min	
Broccoli (florets)	3 min	Steam blanch: 5 min
Brussels sprouts	3–5 min	Varies by size
Cabbage (shredded)	1.5 min	
Carrots (sliced ¼ inch)	2 min	Whole baby: 5 min
Corn (on cob)	7–11 min	Cut from cob: 4 min
Onions (rings)	10–15 seconds	Pieces: 3–7 min
Peas	1.5 min	
Spinach / greens	2 min	
Summer squash (sliced)	3 min	
Winter squash	Cook fully first, then freeze	

Fruits: Most freeze well raw without blanching. Toss sliced fruits in ascorbic acid solution (1 tsp per 2 cups water) or sugar (¾ cup per quart of fruit) to prevent browning. For berries: spread in a single layer on a baking sheet first and freeze until solid before packing in bags — this prevents them from freezing into a solid mass.

Headspace in rigid containers: Leave ½ inch for pints, 1 inch for quarts. Liquids expand when frozen.

Storage Duration at 0°F

Product	Duration
Fruits and vegetables (blanched)	8–12 months
Poultry	6–9 months
Fish and seafood	3–6 months
Ground meat	3–4 months
Whole cuts of beef or pork	6–12 months
Cooked beans or legumes	3–6 months
Tomato sauce or puree	3–6 months
Herb pesto	3–6 months

Frozen food is safe indefinitely at 0°F. Duration guidelines describe quality decline, not safety threshold.

Thawing safely: In the refrigerator (best option; plan ahead). Under cold running water in a sealed bag. As part of cooking without thawing first. Never on the counter at room temperature — the exterior warms into the danger zone while the interior remains frozen.

Method 3: Lacto-Fermentation

Lacto-fermentation is the oldest food preservation method in human history, present in every agricultural culture. It requires no heat, no electricity, and no equipment beyond a jar and salt. The science is straightforward: beneficial bacteria (primarily Lactobacillus species) produce lactic acid as a byproduct of consuming vegetable sugars. The resulting acidic environment — below pH 4.6 — is inhospitable to pathogens including Clostridium botulinum.

A properly fermented product is not just preserved — the acidity actively protects it.

What it works for:
Cabbage (sauerkraut), cucumbers (fermented pickles), carrots, beets, green beans, radishes, turnips, garlic, peppers, cauliflower, kimchi vegetables, hot sauce base.

What it does not work for:
High-sugar fruits ferment into alcohol rather than sour preservation. Soft vegetables deteriorate too quickly. Tomatoes ferment but soften rapidly — better to can or freeze them.

Salt Ratios

Get this right and fermentation works reliably. Get it wrong and you either get mold or inedibly salty food.

2% salt by weight is the standard ratio: 20 grams of pickling salt per 1 kilogram of vegetables, or approximately 1 tablespoon of pickling salt per 1¾ pounds.

A kitchen scale is not optional for fermentation. Volume measurements for salt vary significantly between brands and grind sizes — weight is the only accurate measurement.

For brine (used when vegetables don't produce enough liquid on their own, or when fermenting whole vegetables like cucumbers):
Dissolve 1½ tablespoons pickling salt per quart of water. This makes approximately a 2.5% brine — standard for fermented pickles. For firmer pickles, use up to 3 tablespoons per quart (5%).

Salt type: Non-iodized only. Iodine inhibits fermentation bacteria. Use pickling salt, canning salt, or pure kosher salt (check the label for additives). Do not use table salt with anti-caking agents (clouds the brine, may affect fermentation).

Basic Process (Sauerkraut as Template)

1. Shred or chop vegetable to uniform thickness (¼ inch for cabbage)
2. Weigh the prepared vegetable
3. Apply 2% salt by weight; mix thoroughly
4. Massage or squeeze with clean hands for 5–10 minutes until significant liquid releases
5. Pack firmly into jar; brine should rise above vegetables as you pack
6. If brine does not cover vegetables within 30 minutes, add prepared brine (1½ tablespoons pickling salt per quart water) until covered
7. Weight vegetables down so all surfaces remain below the brine line (a zip-lock bag filled with extra brine works well as a weight — it conforms to the jar and seals off the surface from air)
8. Cover with a cloth or loose lid — fermentation produces CO_2 that needs to escape; do not seal airtight without an airlock
9. Place at room temperature (65–75°F) out of direct sunlight

Timeline: At 70–75°F, most vegetables are fully fermented in 3–4 weeks. At 60–65°F, 5–6 weeks. Below 60°F, fermentation may stall. Above 75°F, fermentation moves faster but texture suffers. Taste weekly after the first week — pull it when you like the flavor.

What's Normal vs. What's Wrong

Normal:

- Bubbling and fizzing, especially in the first week (CO_2 production)
- Brine becoming slightly cloudy
- Sour smell intensifying over time
- White flat film on the surface (Kahm yeast — harmless; skim it off and ensure vegetables are submerged)

Discard the batch:

- Fuzzy mold below the brine surface (pink, black, blue, or green)
- Slimy texture throughout the entire batch
- Rotten odor — distinct from sour fermented smell; trust your nose

Storage After Fermentation

Move to cold storage (refrigerator or cool cellar below 45°F) when fully fermented to the desired flavor. Cold dramatically slows continued fermentation. A fully fermented, well-acidified product refrigerated properly keeps **several months to a year**.

For shelf-stable storage, fermented vegetables can be water-bath canned following NCHFP times — this halts fermentation and produces a shelf-stable product, though it also kills the live cultures.

Method 4: Vinegar Pickling

Vinegar pickling uses acetic acid to achieve the same safety result as lacto-fermentation (pH below 4.6) immediately, without waiting for fermentation to develop. It's faster, more predictable, and the flavor is sharper and less complex than lacto-fermented products.

What it works for:
Cucumbers, peppers, beets (pickled beets are a water bath safe product), green beans (dilly beans), asparagus, cauliflower, carrots, onions, garlic, green tomatoes, watermelon rind.

The One Rule That Cannot Be Broken

Always use **5% acidity (50 grain) vinegar**: standard white distilled vinegar or cider vinegar sold at grocery stores. Do not use homemade vinegar (unknown acidity). Do not use rice vinegar, wine vinegar, or balsamic (lower acidity). Do not reduce the vinegar proportion in a tested recipe.

The safety of a vinegar pickle depends entirely on the final pH of the food being below 4.6. The ratio of vinegar to water in the brine is what achieves this. Changing that ratio — even slightly — can push the final pH above 4.6 into botulism risk territory.

Always use tested recipes for vinegar pickles. The USDA and NCHFP have published tested ratios. Use them exactly.

Standard Dill Pickle Brine (NCHFP Tested)

Yields brine for approximately 7–9 pints:

- 1½ quarts white distilled vinegar (5% acidity)
- ½ cup canning/pickling salt
- ¼ cup sugar (optional)
- 2 quarts water

For each pint jar: 1 dill head (or 1 tsp dill seed), 1 clove garlic, 1/8 tsp crushed red pepper (optional).

Cucumber preparation: Cut 1⁄16 inch off the blossom end of every cucumber before packing (the blossom end contains enzymes that cause softening). Soak cucumbers in ice water for 2–4 hours before packing for best crispness. Use pickling cucumbers, not slicing varieties — slicing cucumbers have thinner skins and softer flesh.

Pickling Process

1. Prepare brine; bring to a boil
2. Pack spices and prepared vegetables into hot jars
3. Pour hot brine over, maintaining ½ inch headspace
4. Remove bubbles; wipe rims with clean damp cloth; apply lids fingertip-tight
5. Process in boiling water bath

Processing times (boiling water bath, dill pickles):

Jar Size	0–1,000 ft	1,001–6,000 ft	Above 6,000 ft
Pints	10 min	15 min	20 min
Quarts	15 min	20 min	25 min

Refrigerator pickles (no canning required): Mix vegetables with brine; refrigerate; consume within 2–3 months. Not shelf-stable.

Beet pickles: Beets are a low-acid vegetable. Pickled beets with adequate vinegar become safe for water bath canning. Never water bath can plain beets without acidification — they require pressure canning.

Common Pickling Mistakes

Soft pickles: Blossom end not removed; processing temperature too high; cucumbers too old at harvest (more than 24 hours from vine to brine is too long).

Cloudy brine: Table salt with anti-caking agents used instead of pickling salt; or normal mineral content in well water. Cloudy brine in vinegar pickles is not necessarily a safety issue, but any cloudiness combined with off odor or bulging lid is.

Hollow cucumbers: Picked too late or overgrown — cucumber seeds developed fully and left a hollow center.

Method 5: Water Bath Canning

Water bath canning seals food in glass jars and processes them in boiling water to destroy pathogens and create a vacuum seal that prevents recontamination. It is safe **only for high-acid foods** — those with a pH below 4.6 before processing.

What it works for:
Fruits (all), fruit juice, jams, jellies, marmalades, properly acidified tomatoes, vinegar-pickled vegetables, salsa (from tested recipes), fermented vegetables (optional second step), fruit butters.

What it absolutely does not work for:
Low-acid vegetables (beans, corn, carrots, beets without added acid, peas, potatoes, peppers without acidification), meats, poultry, fish, soups, or any mixed dish with low-acid components. Using water bath canning for low-acid foods is the primary cause of home canning botulism. If you're canning anything from this list, use a pressure canner — no exceptions.

Equipment Setup

A boiling water canner is a large pot (16+ quart) with a rack that keeps jars off direct heat and allows water to circulate underneath. The pot needs to be deep enough to have 1–2 inches of boiling water above the top of the jar lids during processing.

Fill the canner with water and bring to a simmer (180°F) while you prepare your food. You'll add the filled jars to already-simmering water, then bring it to a full boil to begin timing.

Only use Mason-style canning jars (Ball, Kerr, Bernardin, Golden Harvest). Inspect every jar rim for chips before use — even a hairline chip prevents proper sealing. Use new flat lid discs every time; bands are reusable if undamaged.

The Canning Process, Step by Step

1. Wash jars in hot soapy water or run through dishwasher; keep hot until filling
2. Sterilize jars if processing time is under 10 minutes: submerge empty jars in boiling water for 10 minutes
3. Prepare food per tested recipe
4. Fill jars using a canning funnel, maintaining specified headspace:

- Jams and jellies: ¼ inch
- Fruits: ½ inch
- Tomatoes: ½ inch
- Pickles: ½ inch

5. Remove air bubbles by running a thin spatula around the inside edge of the jar
6. Wipe jar rims with a damp clean cloth — any residue on the rim prevents sealing
7. Apply lids: place flat disc; screw band to fingertip-tight only. Do not over-tighten.
8. Lower jars into canner using jar lifter; do not tilt
9. Ensure 1–2 inches of boiling water above lids; add more boiling water if needed
10. Cover canner; bring to full rolling boil
11. Start timing once a full rolling boil resumes
12. Maintain boil for the entire processing time
13. When time is complete: turn off heat; remove lid; wait 5 minutes
14. Remove jars straight up without tilting; place on towel 1 inch apart to cool
15. Do not press lids or tighten bands — let jars seal naturally as they cool
16. After 12–24 hours: check seals. The lid center should be concave and should not flex when pressed. Any jar that failed to seal: refrigerate and use within a week.

Processing Times (Sea Level)

Food	Jar Size	Time
Applesauce	Pints	15 min
Applesauce	Quarts	20 min
Peaches, hot pack	Pints	20 min
Peaches, hot pack	Quarts	25 min
Tomatoes, whole, raw pack (acidified)	Pints or Quarts	85 min
Tomatoes, whole, hot pack (acidified)	Pints	40 min
Tomatoes, whole, hot pack (acidified)	Quarts	45 min
Dill pickles	Pints	10 min
Dill pickles	Quarts	15 min
Jams and jellies	Half-pints	5 min

Add time for altitude: 1,001–3,000 ft: +5 min. 3,001–6,000 ft: +10 min. 6,001–8,000 ft: +15 min. Above 8,000 ft: +20 min.

Tomatoes: The Special Case

Tomatoes are borderline acid — most varieties sit between pH 4.3 and 4.9. Some lower-acid heirloom varieties may be above 4.6. Always acidify every jar of canned tomatoes:

- Bottled lemon juice: 1 tablespoon per pint, 2 tablespoons per quart. Use bottled, not fresh — fresh lemon juice has variable acidity.
- Citric acid powder: ¼ teaspoon per pint, ½ teaspoon per quart.

Add the acid to the jar before filling. The taste is barely perceptible in the finished product.

Recognizing Spoilage in Canned Food

Before opening:

- Lid that bulges upward (do not open; double-bag and dispose in trash)
- Lid that flexes up and down when pressed (seal failed; discard)
- Visible liquid outside the jar originating from the lid
- Rising air bubbles inside the jar

When opening:

- Spurting liquid when the seal breaks
- Off odors (anything that doesn't smell right)
- Rising bubbles after opening
- Mold on the food surface

For any low-acid canned food showing warning signs: Do not taste. Do not smell closely. Botulinum toxin is colorless and odorless. Treat as potentially contaminated. Dispose of the jar in a sealed heavy bag without opening, or boil the sealed jar for 30 minutes before disposal. Clean any surface that may have contacted the food or liquid with a bleach solution (1 part household bleach to 5 parts water; 30-minute contact time). Wear gloves throughout.

Method 6: Pressure Canning

Pressure canning reaches 240–250°F (116–121°C) — the temperature required to destroy Clostridium botulinum spores in low-acid environments. Boiling water reaches only 212°F, which is insufficient. This is why pressure canning is the only safe method for low-acid vegetables, meats, beans, soups, and mixed dishes.

What it works for:
Green beans, corn, carrots, beets (plain, unpickled), potatoes, peas, peppers (without acidification), all meats and poultry, fish, soups, broths, dried beans, edamame, mixed vegetable dishes.

Never pressure can: Mashed or puréed pumpkin or winter squash (too dense for heat to penetrate to the center — this applies even in a pressure canner; only cube it), dairy products, butter, eggs, oils.

Pressure Canner Setup

Add 2–3 quarts of hot water to the canner as directed by your manufacturer. Load jars on the rack. Lock the lid. Leave the vent open.

The 10-minute vent is mandatory: Heat with the vent open until a steady, uninterrupted stream of steam has been exhausting for 10 continuous minutes. This purges trapped air from the canner. Air in the canner gives a falsely high pressure reading, meaning the actual temperature is lower than the gauge shows — and your food is under-processed. Never skip this step.

After 10 minutes of steady steam: place the weight (or close the petcock). Wait for the canner to reach required pressure. **Start timing only when the canner has reached correct pressure.** Maintain steady pressure throughout — significant fluctuation can cause liquid to be drawn out of jars.

At the end of processing time: turn off heat and let the canner depressurize naturally. Do not run cold water over it or force-release pressure. When the gauge reads 0 PSI (or the weight stops moving), wait an additional 10 minutes, then open the vent, wait another 10 minutes, then open the lid away from you.

Processing Times and Pressures

Pressure figures below are for **dial-gauge canners at 0–2,000 ft elevation** (11 PSI) and **weighted-gauge canners at 0–1,000 ft elevation** (10 PSI).

Vegetable	Jar	Time	Notes
Green/snap beans	Pints	20 min	
Green/snap beans	Quarts	25 min	
Corn, whole kernel	Pints	55 min	Long processing time — plan accordingly
Corn, whole kernel	Quarts	85 min	
Carrots (sliced)	Pints	25 min	
Carrots (sliced)	Quarts	30 min	
Beets (sliced)	Pints	30 min	
Beets (sliced)	Quarts	35 min	
Potatoes (cubed)	Pints	35 min	
Potatoes (cubed)	Quarts	40 min	
Dry beans (presoaked)	Pints	75 min	
Dry beans (presoaked)	Quarts	90 min	
Chicken (boneless)	Pints	75 min	Hot pack
Chicken (with bone)	Quarts	90 min	Hot pack
Ground meat	Pints	75 min	
Ground meat	Quarts	90 min	
Fish (chunks)	Pints	100 min	
Soups (meat and vegetable)	Pints	60 min	See note
Soups (meat and vegetable)	Quarts	75 min	See note

Note on soups: Only process soups for which you have a specific tested recipe. Do not create homemade soup recipes for pressure canning — the density and composition affect heat penetration. NCHFP provides tested soup recipes; use them.

Altitude adjustments (dial-gauge): Add 1 PSI per 2,000 ft above 2,000 ft elevation. **Weighted-gauge:** use 10 PSI at 0–1,000 ft; use 15 PSI above 1,000 ft.

Annual Dial Gauge Calibration

Dial gauges can become inaccurate over time. An inaccurate gauge means you may be processing at lower pressure (and temperature) than you think — and your food is under-processed. Bring your dial-gauge canner lid to your county cooperative extension office before each canning season. Most extension offices will test it for free. If the gauge is off by more than 2 PSI, replace it.

Weighted-gauge canners do not need calibration — the jigger weight is a physical mechanism that releases at the set pressure, regardless of age.

Method 7: Dehydrating and Drying

Dehydration removes the water that microorganisms need to survive. At low enough water activity (Aw), spoilage bacteria, molds, and yeasts cannot grow. It's one of the most ancient preservation methods, and modern electric dehydrators make it reliable and controllable.

What it works for:
Tomatoes (sun-dried style), peppers, herbs, mushrooms, apples, apricots, peaches, plums, figs, berries, corn, green beans (leather britches), potatoes (sliced), bananas, mangoes, jerky.

What it does not work for:
High-moisture vegetables that produce poor texture when dried (cucumbers, lettuce, watermelon). Avocado oxidizes badly. Dairy requires commercial spray-drying. Dense, high-moisture root vegetables work but take a long time and require pre-cooking or long thin slicing.

Temperature: The One Number That Matters

The optimal drying temperature is **140°F.** Going higher cooks the exterior before interior moisture can escape — a condition called case hardening, where a leathery shell traps moisture inside. Case-hardened dried food looks finished but is not dry in the center, and it will mold during storage.

Exceptions:

- Herbs: dry at 95–110°F to preserve volatile oils (the heat-sensitive compounds that make herbs smell and taste like herbs rather than hay)
- Jerky: special considerations — see below

Pretreatments

Vegetables — blanch before drying:
Blanching halts enzyme activity that causes off-colors and off-flavors during drying and storage. See the blanching table in Method 2. Exceptions that do not need blanching: onions, peppers, mushrooms.

Fruits — anti-browning treatment:
Soak cut fruit in ascorbic acid solution (1 teaspoon powdered ascorbic acid dissolved in 2 cups water) for 3–5 minutes, then drain and pat dry. This prevents enzymatic browning without affecting flavor significantly.

Drying Times at 140°F

Times vary significantly based on slice thickness, moisture content, and dehydrator model. These are starting ranges — check frequently toward the end.

Food	Time
Apples (¼ inch slices)	6–12 hours
Apricots (halves)	24–36 hours
Berries (whole)	24–36 hours
Peaches / nectarines (slices)	36–48 hours
Tomatoes (¼ inch slices)	10–18 hours
Peppers (rings or strips)	8–12 hours
Corn (cut kernels)	6–10 hours
Green beans	8–14 hours
Potatoes (¼ inch slices, blanched)	8–12 hours
Mushrooms (slices)	4–8 hours
Herbs (whole sprigs, 95–110°F)	1–4 hours

Doneness test: Vegetables should be leathery to brittle and should shatter or snap when cool, with no moisture when squeezed. Fruits should be pliable and leathery with no moisture beading when pressed, and not sticky. Herbs should crumble completely when rubbed.

Conditioning: The Step People Skip

After drying, pack dried food loosely in a glass jar. Shake once daily for 7–10 days. If moisture beads on the inside of the jar, the food is not fully dry — return it to the dehydrator. This conditioning step detects inadequately dried food before you seal it for long-term storage.

Pasteurization (for outdoor or sun-dried items): Freeze the finished dried food in sealed bags at 0°F for 48 hours, or heat in a 160°F oven for 30 minutes. This kills insect eggs.

Jerky: The Meat Temperature Problem

Dehydrators running at 140°F may not bring meat to an internal temperature of 160°F (the minimum required by USDA for ground meat and poultry jerky) because evaporative cooling keeps the meat's surface temperature lower than the air temperature around it. The dehydrator is processing in the danger zone for an extended period.

Two approaches:

- Post-dry oven finishing: After dehydrating, spread finished jerky on a baking sheet and heat in a 275°F oven for 10 minutes. This brings internal temperature above 160°F without significantly affecting texture.
- Pre-dry steam or roast: Heat raw marinated strips to 160°F before placing in the dehydrator. Steam or roast briefly, then dry. The dehydrator then completes drying rather than needing to pasteurize.

Storage

Store dried food in airtight containers — glass jars with tight-fitting lids, vacuum-sealed bags, or metal cans. Store in a cool, dark location. Light and heat shorten shelf life significantly.

Product	Duration at 60°F	Duration at 80°F
Dried fruit	1 year	6 months
Dried vegetables	6 months	3 months
Dried herbs (whole leaf)	1–2 years	6 months
Jerky (in airtight container)	1–2 months at room temp	Refrigerate for longer

Method 8: Sugar Preservation — Jams, Jellies, and Fruit Butters

Sugar preserves by binding available water — at sufficient concentration (45–55% sugar in the finished product), water activity drops below the threshold for microbial growth. Combined with the natural acidity of fruit and heat processing, this produces a shelf-stable product.

What it works for:
All fruits: strawberries, raspberries, blackberries, blueberries, peaches, apricots, plums, cherries, figs, apples (jelly), grapes (jelly), tomatoes (yes — tomato jam is traditional), citrus (marmalade).

Terminology

Jelly: Made from strained juice only; clear, firm gel.

Jam: Made from crushed or chopped fruit including pulp; soft gel.

Preserves: Large fruit pieces in thick syrup.

Marmalade: Citrus-based with suspended peel.

Fruit butter: Cooked-down smooth purée; no gel; lower sugar than jam.

Gel Point: The Key to Set

Without gel point, jam doesn't set. The gel point is **220°F at sea level** — 8°F above the boiling point of water.

At altitude, the boiling point drops 1°F for every 500 feet of elevation. Subtract accordingly: if you're at 3,000 ft, water boils at approximately 206°F, and your gel point is 214°F.

Without a thermometer: The sheet test. Dip a cold metal spoon into the boiling jam. Hold it horizontally and let jam drip off the edge. When the jam runs together and falls in a sheet rather than individual drops, it has reached the gel point.

Jam with Powdered Pectin (Standard Recipe)

Strawberry jam (NCHFP tested — yields 9–10 half-pints):

- 5½ cups crushed strawberries
- 1 package (1¾ oz) powdered pectin
- 8 cups sugar

Process: Combine fruit and pectin in pot. Bring to a **full rolling boil** — one that cannot be stirred down. Add all sugar at once. Return to a full rolling boil. Boil hard for **exactly 1 minute**. Skim foam from surface. Fill jars to ¼ inch headspace. Process in boiling water bath 5 minutes.

The full rolling boil is not optional. A gentle simmer won't activate the pectin properly and the jam won't set.

Jam Without Pectin

For fruits with sufficient natural pectin (apples, plums, currants, gooseberries, quince) or when using underripe fruit blended with ripe (underripe fruit has more pectin):

Combine fruit and sugar in roughly a 3:2 ratio (3 parts fruit, 2 parts sugar by weight). Bring slowly to a boil, stirring to dissolve sugar. Increase heat; cook at a rapid boil, stirring frequently to prevent scorching, until gel point (220°F). Wide, heavy-bottomed pots reach gel point faster.

Reduced-Sugar Preserves

Reduced-sugar recipes exist and work with no-sugar pectin (Ball RealFruit Low or No-Sugar pectin, Pomona's Universal Pectin). These products have shorter shelf life (6–9 months vs. 18 months for full-sugar), softer set, and different flavor. Follow the manufacturer's instructions exactly — the gel mechanism is different.

Do not simply reduce the sugar in a standard pectin recipe. The sugar is not just flavor — it's part of the preservation and gel chemistry.

Method 9: Salt Curing and Brining

Salt preservation draws moisture out of food through osmosis, raising salt concentration to levels inhospitable to most bacteria, while simultaneously creating flavor through protein chemistry.

What it works for:
Pork (bacon, ham, salt pork), beef (corned beef), fish (salt cod, gravlax), vegetables (for fermentation foundation), olives.

Two Salt Curing Methods

Dry curing: Rub salt (and optionally spices and curing salts) directly onto the surface of the food. Moisture draws out, creating its own brine. Used for bacon, whole hams, cured fish.

Wet brining (immersion): Submerge food in a prepared salt solution. Better penetration into thick cuts. Used for corned beef, poultry brine (as a cooking technique), fermented vegetables.

Salt Ratios by Application

Lacto-fermentation brine (see Method 3): 2% salt by weight.

Pickle brine: 2.5–5% (see Method 4).

Fish — salt cod style:
Apply approximately 1 lb of pure canning salt per 4 lbs of fish (25% salt by weight). Layer fish and salt in alternating layers in a non-reactive container. Refrigerate throughout. Over 2–4 weeks, the fish becomes salt-preserved. Requires soaking in multiple changes of cold water for 24–48 hours before use to rehydrate and remove excess salt. Store at cool temperature (below 50°F) for extended shelf life.

Gravlax (cured salmon):
3 parts canning salt to 2 parts sugar by weight, plus fresh dill and optionally pepper and aquavit. Apply liberally to both sides of the salmon fillet. Wrap tightly in plastic, place in a pan, weight with a heavy pot. Refrigerate 48–72 hours, flipping once daily. Slice thin and serve. This is **not shelf-stable** — refrigerate and use within 5 days after removing from the cure, or freeze.

Curing Salts for Meat: When You Need Them

For any cured meat intended for long storage or smoking at low temperatures, plain salt is insufficient — it doesn't prevent botulism in the anaerobic, protein-rich interior of a thick meat cut.

Prague Powder #1 / Cure #1 / InstaCure #1: Contains 6.25% sodium nitrite. Used for short-cure products that will be cooked or smoked: bacon, ham, corned beef, pastrami, hot dogs. The sodium nitrite prevents C. botulinum and gives cured meat its characteristic color and flavor.

Prague Powder #2 / Cure #2: Contains sodium nitrite + sodium nitrate. Used for long-cure, uncooked dry-cured products with weeks-to-months cure times: traditional dry-cured hams, salami, soppressata.

Standard ratio for Cure #1: 1 oz (28 grams) per 25 lbs of meat for dry cure. Follow package directions precisely and do not improvise — insufficient nitrite doesn't prevent botulism; excessive nitrite is toxic.

The pink color is a safety feature: Curing salts are dyed pink specifically to prevent accidental substitution with plain salt. Do not use plain salt in a recipe calling for curing salt.

Method 10: Smoking

Smoke imparts flavor and deposits antimicrobial compounds on food surfaces. It is **not** a standalone preservation method for home use. Smoking must be combined with salt curing, refrigeration, or freezing. A properly smoked product is a product that has been safely cooked to internal temperature — not one that can sit at room temperature indefinitely.

Hot Smoking: The Standard Home Method

Hot smoking at 225–250°F cooks and flavors food simultaneously.

Required internal temperatures (use a probe thermometer; visual doneness is not safe):

- Poultry (whole or pieces): 165°F
- Whole fish: 145°F internal (at thickest point)
- Pork: 145°F internal + 3-minute rest, or 160°F for greater safety margin
- Beef brisket: usually held at 195–205°F for collagen breakdown, well above safety minimums

Danger zone awareness: Meat sitting between 40°F and 140°F is in the bacterial growth zone. Total time in this range must not exceed 4 hours. During low-and-slow smoking, meat can remain in this zone for extended periods while internal temperature slowly climbs. Use curing salts (Cure #1) for any product that will spend more than 4 hours in the danger zone during the smoking process — particularly cold-smoked items and thick cuts.

Wood selection: Fruitwood (apple, cherry) for poultry and fish; hickory and oak for pork and beef; alder for salmon. Never use treated lumber, pressure-treated wood, pine, or pressed/particle board products.

Cold Smoking: Home Risk Assessment

Cold smoking operates at 68–86°F — temperatures that do not cook the food and that sit within the bacterial growth zone. This is a legitimate traditional technique for products like smoked salmon, but home cold smoking without proper protocols is genuinely high-risk.

If you want to cold-smoke salmon or other fish at home, use a heavily salt-cured product first (3.5%+ salt by weight in the flesh, achieved through a thorough dry or wet cure), and follow a specific tested protocol from a reputable source. Do not cold-smoke improvised recipes.

Storage After Hot Smoking

Hot-smoked products are cooked food, not preserved food. Refrigerate within 2 hours. Consume within 3–4 days. Freeze for longer storage (1–3 months). There is no room-temperature shelf life for hot-smoked products.

Method 11: Alcohol Preservation

High-proof alcohol preserves food by creating an environment inhospitable to microbial growth. This works for fruits, vanilla, citrus, herb tinctures, and flavored spirits — not for meats, vegetables, or anything requiring a lower water activity than alcohol can provide.

The minimum threshold: 40% ABV (80 proof). Wine, beer, and hard cider cannot preserve food reliably without refrigeration. Use 80-proof or higher spirits: brandy, vodka, rum, or whiskey.

Brandied Fruit (Basic Technique)

1. Use ripe, unblemished fruit; wash and dry thoroughly
2. Pack into a clean jar — standard Mason jar works fine; no special processing
3. Optional: add sugar (½ to equal weight of fruit) for sweeter preserves or to make a liqueur
4. Pour 80-proof spirit to completely cover the fruit; no air space above the liquid
5. Seal tightly; store in a cool dark location
6. Ready to eat in 4–6 weeks; improves with age up to 12–18 months

The fruit absorbs alcohol; the alcohol absorbs fruit sugars and flavor. What you end up with is both preserved fruit and a flavored spirit.

Rumtopf (Rum Pot): A traditional German method for preserving the summer harvest. Add each fruit as it comes into season: layer fruit and equal weight of sugar into a crock; cover with rum; seal. Continue adding fruits sequentially through the summer (strawberries, cherries, peaches, raspberries, plums, pears). Keep the rum level above the fruit at all times. Open in December.

Signs of Spoilage

- Surface mold (indicates fruit was exposed to air — ensure submersion)
- Bubbling activity (the sugar content of the fruit has begun fermenting, diluting the alcohol below safe threshold — refrigerate and use soon)
- Off odors other than alcohol and fruit

Method 12: Confit

Confit is a cooked preservation method: food is submerged and cooked in fat at low temperature, then stored under the fat. The fat creates an anaerobic seal that prevents aerobic surface mold. This is not raw preservation — the food is fully cooked.

What it works for: Duck legs, pork (rillettes, belly), chicken thighs, garlic, potatoes (as a cooking method).

Process

1. Salt the meat generously (and optionally with curing salt/spices) 24–48 hours in advance; refrigerate during this salt stage
2. Rinse off excess salt; pat dry
3. Submerge in duck fat, lard, or rendered pork fat in a heavy oven-safe pot
4. Heat fat to 200–225°F (a gentle simmer — not a rolling boil; fat should not smoke)
5. Maintain this temperature until meat is fork-tender and has reached minimum safe internal temperature (165°F for poultry, 160°F for pork)
6. Pack hot meat into clean jars or storage containers; pour hot fat over completely, eliminating air pockets
7. Cool to room temperature; refrigerate immediately

Confit is not shelf-stable at room temperature. The protection provided by the fat seal works only in combination with refrigerator temperature (40°F or below). Traditional French larder storage was possible because cellars maintain near-refrigerator temperatures year-round. A modern pantry at room temperature is not a substitute.

Refrigerated under complete fat submersion: **4–6 weeks**. Frozen: **6 months**.

Part Three: Reference Tables

Crop-to-Method Matrix

Crop	Best Methods	Methods to Avoid	Critical Notes
Tomatoes	Water bath canning (acidified), freezing	Water bath without acidification	Always add lemon juice or citric acid; c
Cucumbers	Lacto-fermentation (best texture), vinegar	Freezing, pressure canning	Remove blossom end; use pickling varieti
Green beans	Pressure canning, freezing (blanched)	Water bath canning without acid	20 min pints / 25 min quarts at 11 PSI
Corn	Pressure canning, freezing (blanched)	Water bath canning without acid	55 min pints / 85 min quarts at 11 PSI;
Carrots	Root cellar (best), pressure canning, fr	Water bath canning without acid	Cold + moist at 32–40°F, 90% humidity; 4
Beets	Root cellar, water bath canning (pickled)	Water bath canning plain without acid	Pickled beets = water bath safe; plain b
Potatoes	Root cellar (primary), pressure canning	Water bath canning, freezing raw	Dark, cold, 38–40°F; cure 10–14 days fir
Sweet potatoes	Root cellar, pressure canning	Refrigerator, water bath, freezing raw	Cure 80°F, 7–10 days; store at 55–60°F m
Winter squash	Root cellar (primary), pressure canning	Pressure canning puréed (not approved)	Cure 80–85°F, 10–14 days; 50–55°F storag
Onions	Root cellar (primary), dehydrating, pick	Oil immersion of fresh (botulism risk)	Cure fully before storage; 32–40°F, low
Garlic	Root cellar, dehydrating, pickling, lact	Oil immersion at room temp (botulism ri	Cure fully; store at 32–40°F (hardneck)
Peppers	Dehydrating (best), freezing, vinegar pi	Water bath without acid	Roasted peppers require pressure canning
Apples	Water bath canning, root cellar, dehydra	—	Store away from all vegetables (ethylene
Berries (all)	Freezing (flash-freeze first), water bat	Root cellar	Freeze individually on tray before bulk
Stone fruits (peaches, plums, apricots)	Water bath canning, freezing, dehydratin	—	Hot pack preferred; ascorbic acid preven
Pears	Water bath canning, dehydrating, root ce	—	Add lemon juice during processing

Herbs	Dehydrating (best), freezing in oil (ba	Oil immersion of fresh at room temp	Harvest before flowering; dry at 95–110°
Pork and beef	Pressure canning, freezing, salt curing	Water bath canning, room-te	All storage cured/smoked meat requi
Poultry	Pressure canning, freezing, salt curing	Water bath canning	165°F minimum internal temperature for a
Fish	Pressure canning, freezing, salt curing	Oil immersion (botulism risk), water bat	Cold smoking requires tested protocols w
Dried beans	Pressure canning (rehydrated), freezing	—	Dry beans store 1–2 years in airtight co

Storage Duration Quick Reference

Method	Typical Duration
Root cellar — root vegetables	4–6 months
Root cellar — squash/sweet potatoes	3–6 months
Root cellar — onions and garlic	6–12 months
Root cellar — apples	2–5 months (variety-dependent)
Frozen vegetables and fruit	8–12 months
Frozen meat (whole cuts)	6–12 months
Frozen ground meat	3–4 months
Frozen fish	3–6 months
Water bath canned (high-acid)	12–18 months
Pressure canned (low-acid)	12–18 months
Lacto-fermented (refrigerated)	Several months to 1 year
Dehydrated fruit (at 60°F)	1 year
Dehydrated vegetables (at 60°F)	6 months
Dried herbs (whole leaf)	1–2 years
Jerky (room temp, airtight)	1–2 months
Vinegar pickles (water bath canned)	12–18 months
Jam and jelly (water bath canned)	12–18 months
Brandied / alcohol-preserved fruit	1–2 years
Confit (refrigerated under fat)	4–6 weeks
Salt-cured fish (refrigerated)	Days to weeks depending on cure level
Hot-smoked meat or fish	3–4 days refrigerated

Critical Safety Reference

The pH 4.6 Rule

Clostridium botulinum cannot produce toxin in environments with pH below 4.6. Water bath canning is safe only for foods that maintain pH below 4.6 throughout processing. Fermentation creates this acidity chemically. Vinegar pickling achieves it immediately with added acid. Any food above pH 4.6 requires pressure canning to reach the 240–250°F needed to destroy spores.

Temperature Thresholds

Temperature	What It Means

250°F (121°C)	Pressure canning maximum — destroys C. b
212°F (100°C)	Boiling water — sufficient for high-acid
180–185°F	Low-temperature pasteurization — pickles
165°F	Required internal temperature for poultr
160°F	Required internal temperature for ground
145°F	Required internal temperature for whole
140°F	Maximum dehydrator temperature to avoid
95–110°F	Dehydrator temperature for herbs
40°F	Refrigerator temperature — slows but doe
0°F	Freezer temperature — stops (does not ki

Botulism in Oil

Fresh garlic or fresh herbs submerged in oil at room temperature = high botulism risk. The combination of low acid, anaerobic environment, and moisture is exactly what C. botulinum needs. Every documented outbreak from home oil-packed garlic has been from room-temperature storage. Refrigerate and use within 4 days, or freeze. No exceptions.

Never Taste-Test Suspicious Food

Botulinum toxin is colorless and odorless. A quantity too small to detect by smell or taste can cause serious illness or death. If a jar looks wrong, smells wrong, or shows any sign of compromise — bulging lid, spurting liquid, unexpected foam, off odor — do not taste it. Discard without opening, or boil sealed in water for 30 minutes before disposal.

If You Are Uncertain, It Isn't Worth the Risk

When in doubt, throw it out. The contents of a quart jar are worth less than what treating foodborne botulism costs.

Appendix: Where to Find Tested Recipes

Tested home canning recipes are not a creative project. The processing times, ratios, and pressures have been established through laboratory testing under specific conditions. Deviating from tested recipes — changing proportions, adding extra vegetables, substituting ingredients — invalidates the safety testing.

Primary sources for safe, tested recipes:

- National Center for Home Food Preservation (NCHFP) — nchfp.uga.edu — the gold standard for home preservation research in the United States; affiliated with University of Georgia; all recipes are USDA-approved
- USDA Complete Guide to Home Canning — available as a free PDF download from USDA; the foundational reference document
- Ball Blue Book Guide to Preserving — the most widely available printed reference; recipes tested and approved; updated periodically
- Your state's cooperative extension service — each state land-grant university operates an extension service with locally relevant guidance; most have free publications and master food preserver programs

Do not use as primary safety references: general food blogs, grandmothers' recipes from before 1994 (when USDA updated processing times), cookbooks without explicit USDA or NCHFP citation, social media canning content, or any recipe that uses open-kettle canning, oven canning, or recommends skipping water bath processing.

Related Seedwarden guides:

- Fermented & Preserved Harvest Handbook — small-batch lacto-fermentation for apartment growers
- Seed Saving Field Manual — closing the loop from harvest to next year's seed
- Survival Garden Regional Plans — what to grow before you preserve it

More from Seedwarden

If you found this guide useful, you might also like:

- Fermented & Preserved Harvest Handbook ($13) — Small-batch lacto-fermentation and live-culture preserving for apartment kitchens
- Meat, Fish & Animal Products Preservation Field Manual ($18) — Smoking, curing, and salt-preserving for protein alongside your plant harvest
- Survival Garden Regional Plans ($18) — Grow exactly what you need to fill a full year of preserved food storage

Browse all Seedwarden guides at [our Etsy shop].